

SRI LANKA INSTITUTE OF ADVANCED TECHNOLOGICAL EDUCATION



Higher National Diploma in Information Technology

Advanced Technological Institute Dehiwala

Individual Project Proposal

Web Based AI-Powered Internship & Job Matching Platform

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Statement of Originality

I hereby declare that the project entitled “Web Based AI-Powered Internship & Job Matching Platform” submitted to IT Department of Advanced Technological Institute Dehiwala is an original work and has not been submitted previously in its entirety or in part to any other institution or university for any degree or diploma. All the information in this document has been obtained and presented in accordance with academic rules and ethical conduct. I have acknowledged all sources of information, ideas, and contributions from others in the text, figures, and tables of this document. I also certify that any assistance received in preparing this project and all sources of information have been acknowledged.

Signed,

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DEH/IT/2023/F/0070

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1. Introduction

Finding suitable internships has become a major challenge for many students in Sri Lanka. Internship experience is important because it allows students to apply their academic knowledge in real-world environments and gain professional skills before entering the job market. However, many students do not know how to find internships that match their knowledge, skills, and career interests.

Currently, most students search for internships through general job portals, social media platforms, or personal connections. These methods are not always effective because they do not provide intelligent guidance about which jobs are suitable for a student's skill level. As a result, students often apply for positions that do not match their qualifications, which leads to rejection and wasted time.

Another problem is that students are not aware of the skills they need to improve in order to obtain their desired internship. Traditional job portals simply display job listings but do not analyze student profiles or provide suggestions for improving employability.

With the advancement of Artificial Intelligence, it is possible to analyze resumes, identify skills, and recommend suitable job opportunities automatically. AI-based systems can also identify missing skills and suggest learning resources to help students improve their career readiness.

Therefore, this project proposes the development of a **web-based AI-powered internship and job matching platform** that connects students with companies offering internship opportunities. The system will analyze student resumes, extract relevant skills, and recommend suitable internships using intelligent matching techniques. In addition, the platform will provide skill gap analysis and learning recommendations to support student career development.

This system aims to simplify the internship search process and provide valuable guidance to students who are preparing to enter the professional workforce.

2. Problem Statement

Many students face significant difficulties when searching for internship opportunities. One major problem is the lack of a centralized platform that intelligently matches student skills with available internships. Students often rely on multiple websites, social media posts, or personal recommendations to find opportunities, which can be inefficient and confusing.

Another problem is that students do not clearly understand whether their skills meet the requirements of a particular internship position. As a result, students may apply for jobs that require skills they do not yet possess. This leads to low success rates and discouragement among students.

Companies also face challenges when recruiting interns because they receive a large number of applications from candidates who may not meet the required skill criteria. This increases the time and effort required for screening applicants.

Therefore, there is a need for a system that can automatically analyze student profiles and job descriptions to provide intelligent matching between candidates and internship opportunities. Such a system can help students identify suitable positions and help companies find candidates whose skills match their requirements.

3. Objectives

General Objective

The main objective of this project is to develop a web-based AI-powered platform that connects students with internship opportunities and provides intelligent recommendations based on their skills and qualifications.

Specific Objectives

- To develop a student profile system that allows users to enter their education, skills, and experience.
- To implement a resume upload feature that allows the system to analyze student CVs.
- To extract relevant skills from resumes using Natural Language Processing techniques.
- To recommend internship opportunities based on the similarity between student skills and job requirements.
- To identify skill gaps between student profiles and desired job positions.
- To recommend learning resources that help students improve their missing skills.
- To allow companies to post internship opportunities and search for suitable candidates.
- To provide an application tracking system for students to monitor their internship applications.

4. Scope of the System

In Scope

The system will provide several features to support both students and companies. Students will be able to create personal profiles, upload their resumes, and view recommended internships based on their skills. The system will analyze resumes using AI techniques to extract important information such as technical skills, education, and experience.

Companies will be able to register on the platform and post internship opportunities with detailed job descriptions and skill requirements. The system will automatically compare these requirements with student profiles to generate match scores.

Students will also receive feedback about missing skills and suggestions for learning resources that can help them improve their employability. Additionally, students will be able to track the status of their internship applications.

Out of Scope

The system will not include payment systems or premium subscription features. The platform will not guarantee job placement or directly manage recruitment decisions made by companies. Mobile application development will also not be included in this project since the system will be implemented as a web-based application.

5. Proposed Solution

The proposed solution is a **web-based internship and job matching platform** that integrates Artificial Intelligence techniques to provide personalized recommendations to students. The system will allow students to create profiles, upload resumes, and receive internship recommendations based on their skills.

The platform will also provide a feature for companies to post internship opportunities and search for candidates whose profiles match their requirements. AI algorithms will analyze both resumes and job descriptions to calculate a match score between students and available internships.

Additionally, the system will provide skill gap analysis by comparing student skills with job requirements. Based on this analysis, the platform will suggest learning resources that help students develop the required skills.

This solution will improve the internship search process by making it more intelligent, efficient, and user-friendly.

6. Methodology

The development of this system will follow the **Agile development methodology**, which allows continuous development, testing, and improvement of the system. Agile methodology is suitable for this project because it allows flexibility and incremental development of system features.

The project will be developed in several stages, including requirement analysis, system design, implementation, testing, and deployment. Each module of the system will be developed and tested separately before being integrated into the final system.

Regular testing and feedback will be conducted to ensure that the system functions correctly and meets the project objectives.

7. Technologies to be Used

The system will be developed using modern web development technologies.

The frontend of the system will be developed using **React** to create an interactive user interface.

The backend server will be developed using **Node.js** and **Express.js**, which will handle application logic, authentication, and communication with the database.

The system will store data using **MongoDB**, which is suitable for handling flexible data structures such as user profiles and job postings.

The AI module will be implemented using **Python** together with **spaCy** and **scikit-learn** to analyze resumes and calculate job matching scores.

Development and coding will be performed using **Visual Studio Code**.

8. Expected Outcomes

The proposed system is expected to significantly improve the internship search process for students. By using AI-based analysis, the system will provide personalized internship recommendations that match student skills and qualifications.

Students will be able to identify skill gaps and receive suggestions for improving their knowledge through recommended learning resources. This will help students better prepare for their desired career paths.

Companies will also benefit from the platform because they will be able to identify suitable candidates more efficiently. Overall, the system will provide a more organized and intelligent approach to internship recruitment.

9. Target Users

The primary users of this system will be students who are searching for internship opportunities and want to improve their career prospects.

Companies and organizations that offer internship opportunities will also use the platform to post job openings and search for potential candidates.

In addition, system administrators will manage the platform, monitor user activities, and ensure that job postings follow the platform guidelines.

10. Time Line

Week 1–2	Project proposal preparation and approval
Week 3–4	Requirement analysis and system design
Week 5–6	Backend development and database design
Week 7–8	Frontend development
Week 9–10	Implementation of AI features
Week 11	System integration and testing
Week 12	Final report preparation and project submission

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